

Objectives

- List the features of ES6
- Explain JavaScript let
- Identify the differences between var and let
- Explain JavaScript const
- Explain ES6 class fundamentals
- Explain ES6 class inheritance
- Define ES6 arrow functions
- Identify set(), map()

In this hands-on lab, you will learn how to:

- Use map() method of ES6
- Apply arrow functions of ES6
- Implement Destructuring features of ES6

Prerequisites

The following is required to complete this hands-on lab:

- Node.js
- NPM
- Visual Studio Code

Notes

Estimated time to complete this lab: **60 minutes.**

Create a React Application named “cricketapp” with the following components:

1. ListofPlayers
 - Declare an array with 11 players and store details of their names and scores using the map feature of ES6

```
return(  
  players.map((item)=>  
    {  
      return(  
        <div>  
          <li>Mr. {item.name}<span> {item.score} </span></li>  
        </div>  
      )  
    }  
  ))  
)
```

- Filter the players with scores below 70 using arrow functions of ES6.

```
return(
  players.map((item)=>
  {
    if(item.score<=70)
    {
      players70.push(item);
    }
  }
  ),
```

2. IndianPlayers

- Display the Odd Team Player and Even Team players using the Destructuring features of ES6

```
export function OddPlayers([first,, third,,fifth]) {
  return(
    <div>
      <li> First : {first} </li>
      <li> Third : {third} </li>
      <li> Fifth : {fifth}</li>
    </div>)}
}
```

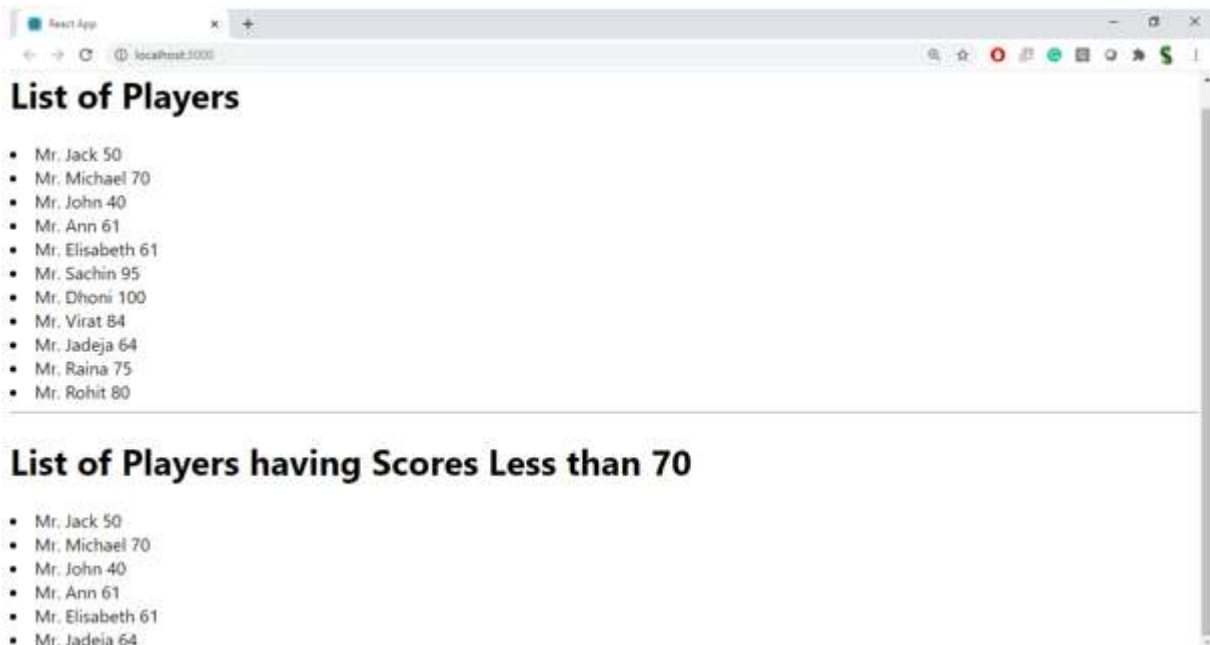
- Declare two arrays T20players and RanjiTrophy players and merge the two arrays and display them using the Merge feature of ES6

```
const T20Players=['First Player','Second Player','Third Player'];
const RanjiTrophyPlayers=['Fourth Player','Fifth Player','Sixth Player'];
export const IndianPlayers=[...T20Players, ...RanjiTrophyPlayers]
```

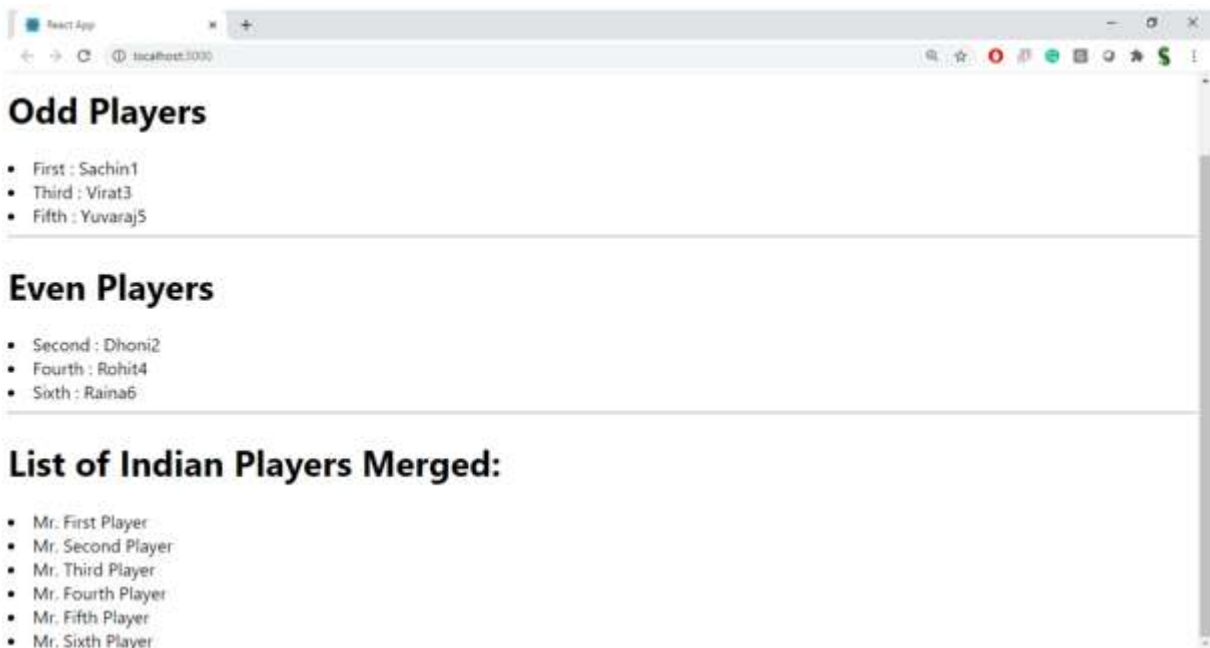
Display these two components in the same home page using a simple if else in the flag variable.

Output:

When Flag=true



When Flag=false



Hint:

```
var flag=true;
if(flag===true)
{
    return(
        <div>
            <h1> List of Players</h1>
            <ListofPlayers players={players}/>
            <hr/>
            <h1> List of Players having Scores Less than 70 </h1>
            <Scorebelow70 players={players}/>
        </div>
    )
}
else{
    return(
        <div>
            <div>
                <h1> Indian Team </h1>
                <h1> Odd Players </h1>
                {OddPlayers(IndianTeam)}
                <hr/>
                <h1> Even Players</h1>
                {EvenPlayers(IndianTeam)}
            </div>
            <hr/>
            <div>
                <h1> List of Indian Players Merged:</h1>
                <ListofIndianPlayers IndianPlayers={IndianPlayers}/>
            </div>
        </div>
    )
}
```

```
Windows PowerShell
PS C:\Users\anana\Downloads\Digital Nurture\Week 5\React\9. ReactJS-HOL> npx create-react-app cricketapp

Creating a new React app in C:\Users\anana\Downloads\Digital Nurture\Week 5\React\9. ReactJS-HOL\cricketapp.

Installing packages. This might take a couple of minutes.
Installing react, react-dom, and react-scripts with cra-template...

added 1301 packages in 1m

272 packages are looking for funding
  run `npm fund` for details

Installing template dependencies using npm...

added 17 packages, removed 2 packages, and changed 1 package in 6s

271 packages are looking for funding
  run `npm fund` for details

Removing template package using npm...

removed 1 package, and audited 1316 packages in 7s

271 packages are looking for funding
  run `npm fund` for details

28 vulnerabilities (9 low, 6 moderate, 13 high)

Windows PowerShell
PS C:\Users\anana\Downloads\Digital Nurture\Week 5\React\9. ReactJS-HOL> cd cricketapp
PS C:\Users\anana\Downloads\Digital Nurture\Week 5\React\9. ReactJS-HOL\cricketapp> code .
PS C:\Users\anana\Downloads\Digital Nurture\Week 5\React\9. ReactJS-HOL\cricketapp> npm start

> cricketapp@0.1.0 start
> react-scripts start

(node:29320) [DEP0176] DeprecationWarning: fs.F_OK is deprecated, use fs.constants.F_OK instead
(Use `node --trace-deprecation ...` to show where the warning was created)
✓ Something is already running on port 3000.

Would you like to run the app on another port instead? ... yes
(node:29320) [DEP_WEBPACK_DEV_SERVER_ON_AFTER_SETUP_MIDDLEWARE] DeprecationWarning: 'onAfterSetupMiddleware'
eprecated. Please use the 'setupMiddlewares' option.
(node:29320) [DEP_WEBPACK_DEV_SERVER_ON_BEFORE_SETUP_MIDDLEWARE] DeprecationWarning: 'onBeforeSetupMiddlewar
deprecated. Please use the 'setupMiddlewares' option.
Starting the development server...
Compiled successfully!

You can now view cricketapp in the browser.

Local:      http://localhost:3001
On Your Network:  http://192.168.0.87:3001

Note that the development build is not optimized.
To create a production build, use npm run build.

webpack compiled successfully
```

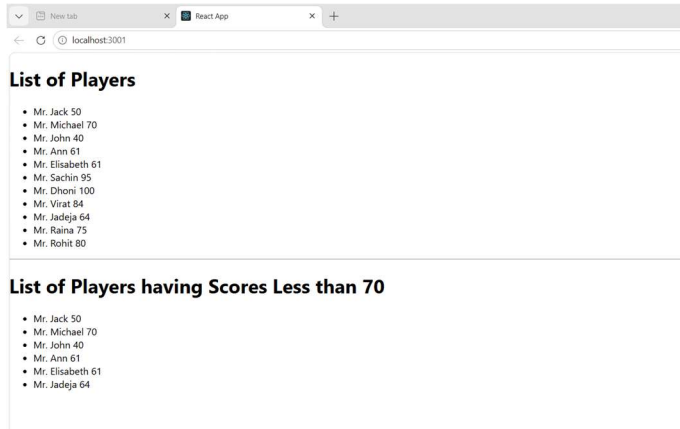
```
JS ListofPlayers.js JS IndianPlayers.js JS App.js X
src > JS App.js > [X] default
1  import {
2    ListofPlayers,
3    Scorebelow70
4  } from "./ListofPlayers";
5
6  import {
7    OddPlayers,
8    EvenPlayers,
9    IndianPlayers,
10   ListofIndianPlayers
11 } from "./IndianPlayers";
12
13 function App() {
14
15   var flag = true;
16
17   if (flag === true) {
18
19     return (
20       <div>
21         <h1>List of Players</h1>
22         <ListofPlayers />
23
24         <hr />
25
26         <h1>List of Players having Scores Less than 70</h1>
27         <Scorebelow70 />
28
29       </div>
30     );
31   } else {
32
33     return (
34
35
```

```
JS ListofPlayers.js JS IndianPlayers.js X JS App.js
src > JS IndianPlayers.js > IndianPlayers
1 import React from "react";
2
3 const T20Players = [
4   "First Player",
5   "Second Player",
6   "Third Player"
7 ];
8
9 const RanjiTrophyPlayers = [
10  "Fourth Player",
11  "Fifth Player",
12  "Sixth Player"
13 ];
14
15 export const IndianPlayers = [
16   ...T20Players,
17   ...RanjiTrophyPlayers
18 ];
19
20 export function OddPlayers([first, , third, , fifth]) {
21
22   return (
23     <div>
24       <ul>
25         <li>First : {first}</li>
26         <li>Third : {third}</li>
27         <li>Fifth : {fifth}</li>
28       </ul>
29     </div>
30   );
31 }
32
33 export function EvenPlayers([, second, , fourth, , sixth]) {
```

src > JS ListofPlayers.js > Scorebelow70

```
1  import React from "react";
2
3  const players = [
4    { name: "Jack", score: 50 },
5    { name: "Michael", score: 70 },
6    { name: "John", score: 40 },
7    { name: "Ann", score: 61 },
8    { name: "Elisabeth", score: 61 },
9    { name: "Sachin", score: 95 },
10   { name: "Dhoni", score: 100 },
11   { name: "Virat", score: 84 },
12   { name: "Jadeja", score: 64 },
13   { name: "Raina", score: 75 },
14   { name: "Rohit", score: 80 }
15 ];
16
17 export function ListofPlayers() {
18   return (
19     <div>
20       <ul>
21         {players.map((item, index) => (
22           <li key={index}>
23             Mr. {item.name} {item.score}
24           </li>
25         ))}
26       </ul>
27     </div>
28   );
29 }
30
31 export function Scorebelow70() {
32
33   const players70 = players.filter(item => item.score <= 70);
34 }
```

VAR FLAG = TRUE



VAR FLAG = FALSE

